

## Chapter 2

# Cellular Automata: Basic Definitions

What is a cellular automaton? This question is not too easy to answer. In the present approach, we define cellular automata in a very narrow sense: the automaton is deterministic and it has a high degree of symmetry. This narrow definition allows us to develop a relatively rich theory. In applications, however, these assumptions are quite often relaxed. Many mathematical systems describe the behavior of single particles, molecules or cells. Such small entities, described on a small spatial scale, do not follow strict deterministic laws but are subject to stochastic variation. Hence it is quite natural to generalize the concept of a cellular automaton in such a way that the local rule depends on random variables.

We have learned from the introduction that a cellular automaton has four components:

- (1) the grid
- (2) the neighborhood
- (3) the states of a single cell
- (4) the local function.

We introduce these notions step by step, and then assemble a cellular automaton from these components.

### 2.1 The Grid

The grid of a cellular automaton can be introduced as a graph. Most applications are based on a square grid in the plane, but also triangular and hexagonal grids and even irregular grids have been used. We define the grid as a graph with symmetries which can be described in terms of a group.

### 2.1.1 Abelian or Regular Grids

The simplest grid starts from a row of points. If this row is infinite in both directions then we can represent the row by the entire numbers. Here we consider the entire numbers as a point set  $\hat{\mathbb{Z}}$ . If the row has only a finite number of points, say  $m + 1$ , then we can identify the first and the last point and obtain a circular point set which we may call  $\hat{\mathbb{Z}}_m$ .

An important property of these rows of points is invariance under a shift. Define  $\sigma : \hat{\mathbb{Z}} \rightarrow \hat{\mathbb{Z}}, z \mapsto z + 1$ , then  $\hat{\mathbb{Z}}$  is invariant under the map  $\sigma$ . We can even use  $\sigma$  to generate  $\hat{\mathbb{Z}}$ , more exactly, to define  $\hat{\mathbb{Z}}$  as being generated by  $\sigma$ . We start with a point  $p \in \hat{\mathbb{Z}}$  and apply all (positive and negative) powers of  $\sigma$ . Apparently here we use  $\mathbb{Z}$  as the additive group of entire numbers. Hence we recover  $\hat{\mathbb{Z}}$  as

$$\hat{\mathbb{Z}} = \{\sigma^n(p) : n \in \mathbb{Z}\}.$$

We can also introduce the group  $G$  generated by the shift  $\sigma$ ,

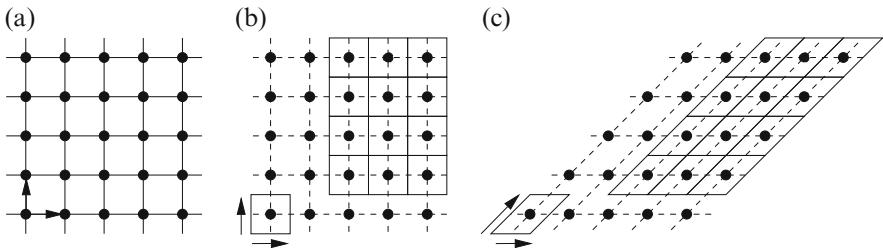
$$G = \{g : g = \sigma^n, n \in \mathbb{Z}\}$$

and write

$$\hat{\mathbb{Z}} = \{g(p) : g \in G\}.$$

Furthermore, we may embed  $\hat{\mathbb{Z}}$  in the real numbers: choose any  $p \in \mathbb{R}$  and define  $\hat{\mathbb{Z}}$  as above. Notice that this construction is very similar to looking at  $\mathbb{R}^d$  as an affine space (a manifold) and also use  $\mathbb{R}^d$  as a vector space of translations of the affine space (tangent space). We define a grid  $\Gamma$  as a graph with  $\hat{\mathbb{Z}}$  as the set of vertices. Two vertices  $x$  and  $y$  are connected by an edge if and only if  $x = \sigma y$  or  $y = \sigma x$ .

This introductory example can be generalized to square grids in the plane or “cubic” grids in  $\mathbb{R}^d$  (see Fig. 2.1a). We start with one point  $p \in \mathbb{R}^d$  and apply shifts  $\sigma_1, \dots, \sigma_d$  that generate a group  $G$  isomorphic to the additive group  $\mathbb{Z}^d$ . If



**Fig. 2.1** Examples of lattice grids: (a) square grid produced by shifting points, (b) square grid and (c) rhombic grid produced by shifting a unit cell. The two translation vectors (corresponding to  $\sigma_1$  and  $\sigma_2$ ) are indicated

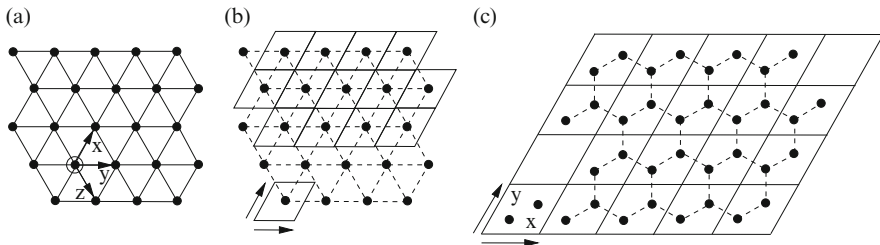
we represent the shifts as vectors in the vector space  $\mathbb{R}^d$  then they are independent. Again the grid is the graph with vertex set  $\hat{\mathbb{Z}}^d = \{g(p) : g \in G\}$  and the property that there is an edge from  $x$  to  $y$  if there is a shift  $\sigma_i$  such that  $x = \sigma_i y$  or  $y = \sigma_i x$  for some  $i \in \{1, \dots, d\}$ .

We have constructed the square grid in three steps, first a set of grid points and then the edges and then the embedding in  $\mathbb{R}^2$ . We can generate it in a quite different way. We define a “unit cell” or “elementary tile” as a subset of  $\mathbb{R}^2$ , for example a square of edge length 1 centered at the origin (see Fig. 2.1b) and use the same generators  $\sigma_1$  and  $\sigma_2$  as before. We see that the plane is tessellated with copies of the elementary cell. A rhombic pattern can be produced by the first approach using a different embedding of the graph into  $\mathbb{R}^2$  or by using a rhombic unit cell and appropriate shift operators, see Fig. 2.1c.

We know that by definition every planar periodic pattern is generated by a “unit cell” and two shift operators. The differences between the two approaches become obvious if we look at a triangular pattern as in Fig. 2.1b. With the first approach we need three generators  $x, y, z$  as in Fig. 2.2a. If we use only two of these we get the same set of grid points but only a rhombic grid. On the other hand we can choose a unit cell, for example a rectangle centered at the origin (see Fig. 2.2b), and use two shift operators. Finally we look at the hexagonal grid in Fig. 2.2c. There is no way to produce the set of grid points as the images of a single point under the action of some shifts. But we can define a rhombic elementary cell as in Fig. 2.2b and produce the grid with two generators.

We will later use the graph structure to indicate the neighborhood of a point. Points within a neighborhood can, but do not have to, interact. However, formally we may choose the neighborhood larger than the sets of points that indeed interact locally. The construction in Fig. 2.2c only allows to distinguish copies of the unit cell, not single points. This may force us to choose a large neighborhood, but is allowed in the framework for cellular automata we develop here.

**Definition 2.1.1** A regular or Abelian grid  $\Gamma$  consists of an elementary tile  $c_0$  and commuting shifts  $\sigma = \{\sigma_1, \dots, \sigma_d\}$  generating a group  $G$ . The grid is determined



**Fig. 2.2** A triangular grid is generated (a) by three shift operators or (b) using a special unit cell and two shift operators. (c) Also a hexagonal grid can be generated by an appropriate unit cell and two shift operators. The generated hexagonal structure is indicated by the dashed lines

by  $G$  and the set of generators  $\sigma$ ,  $\Gamma = \Gamma(G, \sigma)$ . The tiles are given as

$$gc_0, \quad g \in G$$

or as

$$\prod_{i=1}^d \sigma_i^{z_i} c_0, \quad z_i \in \mathbb{Z}.$$

Either  $\sigma_i^z \neq \sigma_i^{z'}$  for  $z \neq z'$ , or there is  $n_i \in \mathbb{N}$  such that  $\sigma_i^0 = \sigma_i^{n_i}$ . In the first case, we call the direction corresponding to the shift operator  $\sigma_i$  unbounded, in the second case bounded.

*Remark 2.1.2*

- (1) Regular grids play a central role in crystallography, where one aims at classifying all regular patterns. In general, the elementary tiles exhibit a certain symmetry. One can show that the symmetry group of the resulting grid in the plane is one of seventeen possible groups, the so-called crystallographic groups of the plane. Also in arts, e.g. in the classification of mosaics, these groups play a role [79]. Similar results hold true for regular grids embedded in spaces of higher dimension.
- (2) We do not need to know the nature of the object  $c_0$ . Although it may be quite useful to have points in  $\mathbb{R}^d$  in mind, this is by no means necessary. Think of the pictures of M.C. Escher who fills his tiles with birds and fishes.

### 2.1.2 Non-Abelian Grids

We started with a regular grid and we obtained an Abelian group which is in some sense equivalent to the grid. Can we also do the converse? Given a group, can we construct a grid, or—more general—a graph? Already in 1878, Arthur Cayley proposed a “graphical representation” of a finitely generated group (find in Appendix A.2 the definition of a finitely generated group) which we now call a Cayley graph.

**Definition 2.1.3** Consider a (not necessarily Abelian) finitely generated group  $G$  with a set of generators  $\sigma = \{\sigma_1, \dots, \sigma_d\}$ . The vertices of the Cayley graph  $\Gamma = \Gamma(G, \sigma)$  are the elements of  $G$ . The vertices  $g_1, g_2 \in G$  are connected by an edge if and only if there is a generator  $\sigma_i \in \sigma$  such that

$$g_1 = g_2 \sigma_i \quad \text{or} \quad g_2 = g_1 \sigma_i.$$

We call the neutral element  $e \in G$  the origin of the Cayley graph. The vertices are called sites or cells of the grid.

*Remark 2.1.4* The Cayley graph is constructed from a group. In general, the Cayley graph depends on the choice of the generators. If different choices are made, then the corresponding Cayley graphs may be not isomorphic. For example, if  $G = \langle \sigma_1, \dots, \sigma_m \rangle$ , then we can add any other element  $\tau \in G$  to the set of generators,  $G = \langle \sigma_1, \dots, \sigma_m, \tau \rangle$ . The Cayley graphs corresponding to  $\sigma_1, \dots, \sigma_m$  and to  $\sigma_1, \dots, \sigma_m, \tau$  are different since the grid contains more edges. However, for a given group, the number of vertices in a Cayley graph is always the same.

*Example 2.1.5* The Cayley graph of a finitely generated Abelian group is a grid (possibly finite).

*Example 2.1.6* Can we visualize the Cayley graph of a non-Abelian group? Yes, we can, and the visualization is used in group theory to find underlying symmetries of a given group. However, the resulting graph need not be planar. Consider the symmetry group of an equilateral triangle, the dihedral group  $D_3$ . Let  $\sigma$  be the rotation by  $2\pi/3$  and let  $\tau$  be one of the reflections (left-right in a standard picture). Then

$$D_3 = \langle \sigma, \tau : \sigma^3 = e, \tau^2 = e, \tau\sigma\tau\sigma = e \rangle = \{e, \sigma^1, \sigma^2, \tau, \tau\sigma, \tau\sigma^2\}.$$

The Cayley graph has six vertices. The table of multiplications from the right can be found in Table 2.1.

The Cayley graph is shown in Fig 2.3. This finite graph shares some features with a torus: there is an origin (the unit element), and two “shift operators”  $\sigma$  and  $\tau$  generating the graph. The origin can be shifted into any point and any point can be shifted into the origin.

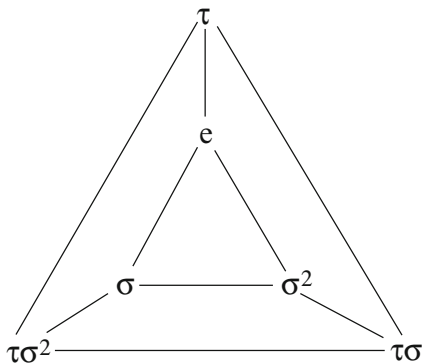
If we are honest, we did not gain much by this example: the topology of this graph resembles that of the regular grid described by the Abelian group  $\mathbb{Z}_3 \times \mathbb{Z}_2$ . In general: A Cayley graph for  $D_n$  resembles a Cayley graph for  $\mathbb{Z}_n \times \mathbb{Z}_2$ . There is only one difference: If we apply the generator  $\sigma$  with  $\sigma^n = e$  then in the Abelian case we visit the vertices in the inner and in the outer circle in the same direction while we visit them in the opposite direction in the case of  $D_n$ . Here we consider undirected graphs, so this difference has no consequences for the theory of cellular automata.

**Table 2.1** Multiplication table (from the right) for the symmetry group of an equilateral triangle

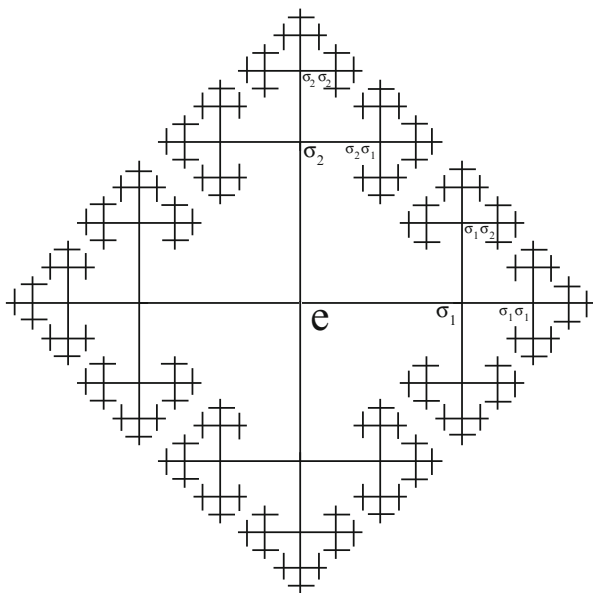
| Vertex         | Multiply with $\sigma$ | Multiply with $\sigma^2$ | Multiply with $\tau$                                 |
|----------------|------------------------|--------------------------|------------------------------------------------------|
| $e$            | $\sigma$               | $\sigma^2$               | $\tau$                                               |
| $\sigma$       | $\sigma^2$             | $e$                      | $\sigma\tau = \tau\sigma^2$                          |
| $\sigma^2$     | $e$                    | $\sigma$                 | $\sigma^2\tau = \sigma\tau\sigma^2 = \tau\sigma$     |
| $\tau$         | $\tau\sigma$           | $\tau\sigma^2$           | $e$                                                  |
| $\tau\sigma$   | $\tau\sigma^2$         | $\tau$                   | $\tau\sigma\tau = \sigma^2$                          |
| $\tau\sigma^2$ | $\tau$                 | $\tau\sigma$             | $\tau\sigma^2\tau = \tau\sigma\tau\sigma^2 = \sigma$ |

**Fig. 2.3** Cayley graph for the group  $D_3$

D refers to directions



**Fig. 2.4** First levels of the Cayley graph for the free group  $\langle \sigma_1, \sigma_2 \rangle$ . The points  $e, \sigma_1, \sigma_2, \sigma_1\sigma_2, \sigma_2\sigma_1, \sigma_1\sigma_1$  and  $\sigma_2\sigma_2$  are indicated. For the complete graph, the branching proceeds to ever finer levels



*Example 2.1.7* Consider the Cayley graph of the free group generated by two symbols,

$$G = \langle \sigma_1, \sigma_2 \rangle .$$

The groups  $\langle \sigma_1 \rangle$  and  $\langle \sigma_2 \rangle$  are isomorphic to  $(\mathbb{Z}, +)$ . At each vertex there are two crossing branches (or four arms) corresponding to the application of  $\langle \sigma_1 \rangle$  and  $\langle \sigma_2 \rangle$  separately. In order to obtain a nice picture we let the scale shrink with distance from the unit element. We obtain a self-similar graph with a  $D_4$  symmetry. The unit element is not distinguished. We can choose any element as the starting point, see Fig. 2.4.

Some properties of regular grids extend to general Cayley graphs.

**Proposition 2.1.8** *Let  $G$  be a finitely generated group. Any Cayley graph of  $G$  is connected. Furthermore, the group action of a group element from the left,*

$$\sigma_g : \Gamma \rightarrow \Gamma, \quad h \mapsto g \cdot h$$

*is an automorphism of the Cayley graph.*

*Proof* Connectedness: Let  $g \in G = \langle \sigma_1, \dots, \sigma_d \rangle$ . The element  $g$  can be represented as

$$g = \prod_{k=1}^l \tau_k$$

with  $\tau_k \in \{\sigma_1, \sigma_1^{-1}, \dots, \sigma_d, \sigma_d^{-1}\}$ . Let  $g_j = \prod_{i=1}^j \tau_i$  with  $g_0$  being the neutral element and  $g_l = g$ . Since  $g_{i+1} = g_i \tau_{i+1}$ , there is a path in the Cayley graph  $e \rightarrow g_1 \rightarrow \dots \rightarrow g_l = g$ . There is a path from any point to the unit element and hence a Cayley graph is connected.

**Automorphism:** First of all, multiplication of  $G$  with a group element is a permutation of  $G$ . Furthermore, since  $g_1 \tau = g_2$  is equivalent with  $h g_1 \tau = h g_2$ , multiplication from the left with a group element neither destroys nor creates edges. Thus, multiplication from the left is an automorphism of the Cayley graph.  $\square$

*Remark 2.1.9*

- (1) For an Abelian grid, e.g.  $\mathbb{Z}^d$ , the operator  $\sigma_g$  corresponds directly to a shift of the complete grid by the vector  $g$ . Also in non-Abelian grids the operators  $\sigma_g$  play the role of shift operators.
- (2) An immediate consequence of Proposition 2.1.8 is that any element  $g$  of the Cayley graph may be moved into the origin (resp. the origin into any given element) via  $\sigma_{g^{-1}} : G \rightarrow G, h \mapsto g^{-1}h$  (resp.  $\sigma_g$ ). This observation corresponds to the fact that in a regular grid there is a tagged origin, but it does not matter where in the grid this origin is located. Via translation this origin may be moved to any point.
- (3) A further immediate consequence is that any vertex (group element) has the same degree, i.e., the same number of adjacent edges. Also this fact is obvious for a regular grid.

**Definition 2.1.10** Since a Cayley graph is connected, a finite distance measure can be introduced. Define the length of any edge as 1. Then the distance  $d_c(g_1, g_2)$  of two elements  $g_1$  and  $g_2$  in the Cayley graph is defined as the length of a shortest path between these two elements. The metric  $d_c$  on the Cayley graph is called the Cayley metric.

In the case of  $\mathbb{Z}^d$ , the Cayley metric coincides with the  $l^1$ -metric for coordinate vectors,

$$d_c(x, y) = \|x - y\|_1 \quad \text{for } x, y \in \mathbb{Z}^d.$$

**Notation** In the following, a grid  $\Gamma$  denotes the Cayley graph of a finitely generated group  $G$ . We call the grid regular or Abelian if this group is Abelian. We will denote the neutral element by  $e$ , in the Abelian case we also use  $0$ . Accordingly, the group operation is either denoted by multiplication, or, if we wish to emphasize the Abelian case, by addition  $+$ .

Note that the finite generated groups are by no means the most general structures that allow to define cellular automata. Cellular automata are e.g. also discussed on free monoids [29, 30]. However, as we will see, in some sense finitely generated groups are sufficient to understand cellular automata completely, at least in a topological sense: any continuous map on a metric Cantor space can be embedded in a cellular automaton on an finitely generated group (see Chap. 3 below). This generality is the main reason, why finitely generated groups as spatial structure are of special interest and discussed in depth, see e.g. the book by Ceccherini-Silberstein and Coornaert [27].

## 2.2 The Neighborhood

One essential property of a cellular automaton is locality: each cell interacts only with a fixed number of neighboring cells. Let a Cayley graph  $\Gamma$  be given.

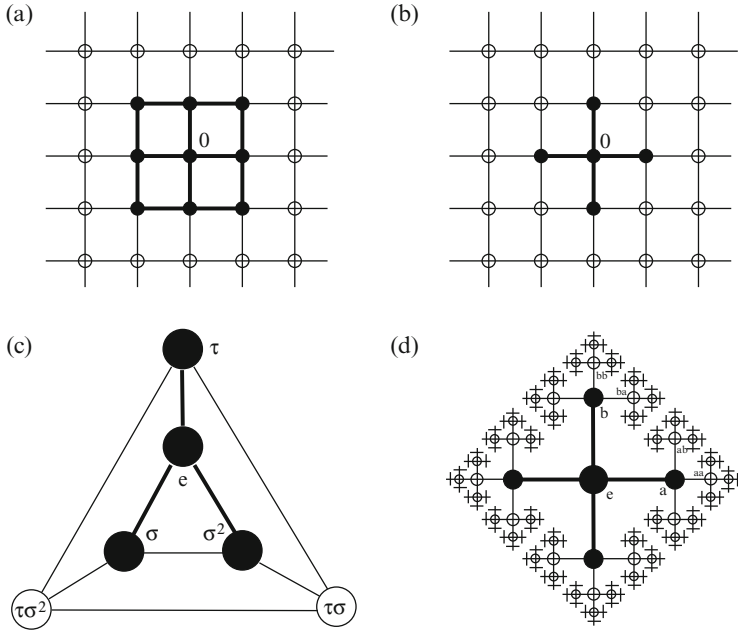
First define the neighborhood  $D_0$  of the origin.  $D_0$  can be any finite set of vertices. The elements of the neighborhood need not be connected to the origin by edges, and the origin itself need not be in its neighborhood. In this generality, the concept of a neighborhood bears little relation to a topological neighborhood.

In order to define the neighborhood of other vertices we use the homogeneity of the Cayley graph: The neighborhood of the point  $g$  is obtained by shifting  $D_0$  to this point.

**Definition 2.2.1** Let  $G$  be a finitely generated group and  $\Gamma = \Gamma(G, \sigma)$  a Cayley graph. Let  $D_0$  be a neighborhood of  $e$  (i.e., any finite set of vertices of  $\Gamma$ ). The neighborhood of a point  $g \in \Gamma$  is given by

$$D_g = \sigma_g D_0 = g D_0.$$

In regular grids we can define certain types of neighborhoods using the distance (Fig. 2.5).



**Fig. 2.5** (a) Moore neighborhood of  $0 \in \mathbb{Z}^2$ . (b) von Neumann neighborhood of  $0 \in \mathbb{Z}^2$ . (c) and (d) the von Neumann neighborhood of the unit element for the group  $D_3$  and the free group  $\langle a, b \rangle$ , respectively. The neighborhood is indicated by *bullets* and *fat edges*

**Definition 2.2.2** For the regular grid  $\mathbb{Z}^d$ , the **von Neumann neighborhood** of a point  $x$  is defined as the set

$$D_x = \{y \in \mathbb{Z}^d : \|y - x\|_1 \leq 1\}$$

and the **Moore neighborhood** is defined as

$$D_x = \{y \in \mathbb{Z}^d : \|x - y\|_\infty \leq 1\}.$$

In this case **translation invariance is evident**.

The concept of a von Neumann neighborhood can be extended to arbitrary Cayley graphs.

**Definition 2.2.3** Let  $G$  be a finitely generated group and  $\Gamma = \Gamma(G, \sigma)$  a Cayley graph with set of generators  $\sigma$ . The **von Neumann neighborhood of the origin** is defined as

$$D_0 = \{g \in \Gamma : d_c(g, e) \leq 1\}$$

i.e., it consists of the immediate neighbors of the origin in the Cayley graph and the origin itself.

By definition, the von Neumann neighborhood of any element  $g$  is obtained by shifting the von Neumann neighborhood to that element. We can also obtain the neighborhood by evaluating distances near  $g$ .

*Remark 2.2.4* The von Neumann neighborhood of an element  $g$  is given by

$$D_g = \{h \in \Gamma : d_c(h, g) \leq 1\}.$$

The Moore neighborhood on the grid  $\mathbb{Z}^2$  cannot be defined by using the distance alone. We can however introduce two more generators and get an extended grid with an increased number of edges. Then the Moore neighborhood of  $\mathbb{Z}^2$  becomes the von Neumann neighborhood of the extended grid. But, unfortunately, the extended grid is not planar. Draw it!

## 2.3 Elementary State and the Global State

Next we introduce the local and global state.

**Definition 2.3.1** Let  $\Gamma(G, \sigma)$  be a Cayley graph. We define functions on  $\Gamma$  with values in a finite set  $E$  which we call the set of elementary or local states, or the “alphabet”. A function

$$u : \Gamma \rightarrow E$$

is called a (global) state. The set  $E^\Gamma = \{u : \Gamma \rightarrow E\}$  is the state space. If  $g$  is a vertex (cell) then  $u(g) \in E$  is called the state of the cell.

We study transformations of  $E^\Gamma$ . Any permutation  $\pi$  of the elements of  $E$  induces an invertible transformation of the set  $E^\Gamma$  by

$$\hat{\pi} : E^\Gamma \rightarrow E^\Gamma, \quad u \mapsto \pi \circ u.$$

In this way we get a group acting on  $E^\Gamma$  which is isomorphic to the group of permutations of  $E$ . A simple example is  $E = \{\text{white}, \text{black}\}$ . Then the non-trivial permutation is the flip which changes black cells into white and conversely white cells into black.

These transformations act on the image of  $u$ . Now consider transformations of the pre-image. An element of the group  $g \in G$  induces a shift isomorphism of the Cayley graph,  $\sigma_g : \Gamma \rightarrow \Gamma, h \mapsto g \cdot h$ . Also the shift operators can be lifted to  $E^\Gamma$  by

$$\hat{\sigma}_g : E^\Gamma \rightarrow E^\Gamma, \quad u \mapsto u \circ \sigma_g.$$

In this case, the states of cells are shifted by the action of the operator  $\sigma_g$ . In the Abelian case the state of a cell is moved by a constant shift vector to another cell on the grid.

The lift  $\hat{\pi}$  commutes with the shift operators  $\hat{\sigma}_g$ ,

$$\hat{\pi} \circ \hat{\sigma}_g = \hat{\sigma}_g \circ \hat{\pi}.$$

If you think about it, this is a basic property of maps: transformations of the image commute with transformations of the pre-image. Since no confusion of  $\sigma_g$  and  $\hat{\sigma}_g$  is possible, we drop the hat again.

**Notation** We write  $\sigma_g : E^\Gamma \rightarrow E^\Gamma$  instead of  $\hat{\sigma}_g : E^\Gamma \rightarrow E^\Gamma$ , as it is always possible to clearly decide if  $\sigma_g$  acts on  $\Gamma$  or on  $E^\Gamma$ .

Sometimes it makes sense to abandon the homogeneous structure of the set  $E$  and introduce a neutral state or resting state  $r$ . In many applications we have  $E = \{0, 1\}$  and the neutral state is 0.

**Definition 2.3.2**  $E$  is called a set of states with a resting state, if there is exactly one marked element  $r \in E$  which is called the resting, quiescent or background state.

Let  $\Gamma$  be a grid. For  $u \in E^\Gamma$  the support is defined as the set of cells  $g \in \Gamma$  with a state different from the resting state,

$$\text{supp}(u) = \{g : u(g) \neq r\}.$$

A state has finite support, if and only if  $|\text{supp}(u)| < \infty$ . A state with finite support is also called a configuration. The set of all configurations is denoted by  $(E^\Gamma)_c$ .

Later, we will, in a slightly abuse of this definition, also call the state of a finite (sub-)graph a configuration. A pattern always indicates the state of a finite (sub-)graph, and a word is a pattern on a one-dimensional graph  $(\mathbb{Z}, \mathbb{Z}_m)$ .

Another important subset of  $E^\Gamma$  are the periodic states. For a regular grid  $\mathbb{Z}^d$  generated by shift operators  $\sigma_1, \dots, \sigma_n$ , a state  $u$  is periodic if for each  $i$  there is some  $k_i \in \mathbb{N}$  such that  $u$  is invariant under  $\sigma_i^{k_i}$ . It is not immediately clear how to generalize this concept to arbitrary Cayley graphs. Yukita [186] proposes the following approach. Given a state  $u$  on a Cayley graph  $\Gamma(G, \sigma)$ , we first determine all group elements  $g \in G$  that correspond to transformations  $\sigma_g : E^\Gamma \rightarrow E^\Gamma$  that leave  $u$  invariant. This set is non-empty as the unit element always has this property. Moreover, these group elements form a subgroup  $H = H(u)$ . In the Abelian case considered before, this subgroup has a finite index which is given by  $\prod_{i=1}^n k_i$  if we choose the numbers  $k_i$  to be minimal. We turn this observation into a definition.

**Definition 2.3.3** Let  $G$  be a finitely generated group,  $\Gamma = \Gamma(G, \sigma)$  a Cayley graph for this group, and  $u \in E^\Gamma$  a state. Define a subgroup of  $G$  by

$$H(u) = \{g \in G : \sigma_g(u) = u\}.$$

The state  $u$  is called  $(H)$ -periodic if the index  $[G : H(u)]$  is finite. The set of all periodic states is denoted by  $(E^\Gamma)_H$ .

Note that we need to know an  $H$ -periodic state only for one representative out of each right-coset to know the state completely, as it is constant on  $Hg$  for all  $g \in G$ . As the index of  $H$  in  $G$  is finite,  $u$  is determined by its value at a finite number of sites.

Another useful concept is to equip the set  $E$  with an algebraic structure, e.g. to consider alphabets that have group or ring structure. If the dynamics respects this algebraic structure, it is possible to understand the dynamics much better than in the general case. An example is again  $E = \{0, 1\}$  which can be seen as  $\mathbb{F}_2$ , the field with two elements. We will return to this aspect in Sect. 10.

## 2.4 The Local and the Global Function

By now we have a grid  $\Gamma$  (a Cayley graph), the elementary states  $E$ , the state space  $E^\Gamma$  and the neighborhood  $D_0$ . Next we introduce the local state space as the set of all functions (local states) from the neighborhood  $D_0$  to the set  $E$ , i.e.,

$$E^{D_0}.$$

For any point  $g \in \Gamma$  we define  $E^{D_g}$  as the local state space at  $g$ .

The local function  $f_0$  is a map from  $E^{D_0}$  to  $E$ ,

$$f_0 : E^{D_0} \rightarrow E.$$

The local function is the central “machine” of a cellular automaton. Knowing the local function, we construct the global function which maps a global state into a global state. The new state of the cell  $g$  is computed as follows: Shift  $D_0$  by  $g$ , then restrict the state  $u$  to  $g(D_0)$ , then apply  $f_0$ .

We define a cellular automaton as follows.

**Definition 2.4.1** Let  $G$  be a finitely generated group,  $\Gamma = \Gamma(G, \sigma)$  a Cayley graph. Let  $D_0$  be a finite subset of  $\Gamma$ , called the neighborhood of the unit element  $e \in G$  and  $E$  a finite set of local states. Let furthermore  $f_0 : E^{D_0} \rightarrow E$ . We define

$$f : E^\Gamma \rightarrow E^\Gamma, \quad u \mapsto f(u)$$

with

$$f(u)(g) = f_0((u \circ \sigma_g)|_{D_0}) \quad \text{for } g \in \Gamma.$$

The tuple  $(\Gamma, D_0, E, f_0)$  is called a cellular automaton. The time course of a state  $u_0 \in E^\Gamma$  (the orbit or trajectory starting with  $u_0$ ) is given by the sequence  $(u_0, u_1, u_2, \dots)$  with  $u_i = f(u_{i-1})$ ,  $i = 1, 2, \dots$

An alternative description, slightly imprecise, reads

$$f(u)(g) = f_0(u|_{g(D_0)}) \quad \text{for } g \in G.$$

The class of cellular automata is very rich and exhibits all kind of behaviors, from very simple, predictable behavior to complex and chaotic dynamics. Only very few theorems apply to the whole class—often enough one considers subclasses. At present we only mention a few classes.

**Definition 2.4.2** A cellular automaton  $(\Gamma, D_0, E, f_0)$  is called a cellular automaton with resting state, if the set  $E$  has a resting state  $r$  and if for  $u_0 \in E^{D_0}$ ,  $u_0(g) = r$  for all  $g \in D_0$ , we have

$$f_0(u_0) = r.$$

A cellular automaton with resting state maps any state with finite support into a state with finite support.

**Proposition 2.4.3** A cellular automaton with resting state leaves the set  $(E^\Gamma)_c$  (forward) invariant,

$$f : (E^\Gamma)_c \rightarrow (E^\Gamma)_c.$$

As  $(E^\Gamma)_c$  is invariant, the restriction of the cellular automaton to this set is well defined.

**Definition 2.4.4** Let  $(\Gamma, D_0, E, f_0)$  denote a cellular automaton, and  $X \subset E^\Gamma$  such that  $X$  is closed w.r.t. all shift operators,  $\sigma_g(X) \subset X$  for all  $g \in G$ . If  $f(X) \subset X$ , we call the restriction of  $f$  to  $X$  a (cellular) sub-automaton.

**Corollary 2.4.5** A cellular automaton with resting state  $(\Gamma, D_0, E, f_0)$  has the sub-automaton  $f|_{(E^\Gamma)_c}$ .

In general, a subautomaton is not an automaton.

We want to show that the set of periodic states is forward invariant. We first prove another simple but important property of the global function: it commutes with the lifts of all shift operators.

**Proposition 2.4.6** Let  $(\Gamma, D_0, E, f_0)$  be a cellular automaton with global function  $f$ . Then,

- (a)  $f$  commutes with the lifts of all shift operators.
- (b) The set of periodic states is invariant under  $f$ .

*Proof*

(a) For  $h, g \in G, u \in E^\Gamma$ , we find

$$\sigma_g \circ f(u)(h) = f(u)(g h) = f_0(u|_{\sigma_{g h}(D_0)}) = f_0(\sigma_g(u)|_{\sigma_h(D_0)}) = f \circ \sigma_g(u)(h).$$

(b) Let  $g \in H(u)$ . Then,  $\sigma_g \circ f(u) = f \circ \sigma_g(u) = f(u)$ , i.e.,  $H(u) \subset H(f(u))$  and hence for the index

$$[G : H(f(u))] \leq [G : H(u)].$$

Thus, the image of a periodic state under a cellular automaton is again periodic.  $\square$

**Corollary 2.4.7** Any cellular automaton  $(\Gamma, D_0, E, f)$  has the sub-automaton  $f|_{(E^\Gamma)_P}$ .

Certain classes of cellular automata have very simple local functions and can be easily analyzed. We introduce some of the most common types.

**Definition 2.4.8** The automaton  $(\Gamma, D_0, E, f_0)$  with  $E = \{0, 1\}$  is called **totalistic** if  $f_0(u)$  depends only on the number of cells occupied, i.e., on the number of “1”,

$$f(u) = f^*(s), \quad s = \sum_{y \in D_0} u(y)$$

with some function  $f^* : \mathbb{N}_0 \rightarrow E$ . The automaton  $(\Gamma, D_0, E, f_0)$  is called **semitotalistic** if  $f_0(u)$  depends only on the value  $u(0)$  and on the sum of the values on  $D'_0 = D_0 \setminus \{0\}$ . The automaton is semi-totalistic if there is a function of two variables  $f^* : E \times \mathbb{N}_0 \rightarrow E$  such that

$$f_0(u) = f^*(u(0), s'), \quad s' = \sum_{y \in D'_0} u(y).$$

Given the automaton  $(\Gamma, D_0, E, f_0)$  with  $E = \{0, 1\}$ , there is a partial order between local states  $u_i \in E^{D_0}$ ,  $i = 1, 2$ , defined by

$$u_1 \leq u_2 \quad \Leftrightarrow \quad \forall g \in D_0 : u_1(g) \leq u_2(g).$$

We introduce the class of automata that respect this order.

**Definition 2.4.9** The automaton  $(\Gamma, D_0, E, f_0)$  with  $E = \{0, 1\}$  is called **monotone** if

$$u_1 \leq u_2 \Rightarrow f_0(u_1) \leq f_0(u_2).$$

## 2.5 Excursion: The Growth Function of a Cayley Graph

In later sections we will use the concept of the growth function of a Cayley graph.

**Definition 2.5.1** Given a Cayley graph  $\Gamma$ , let  $\Gamma_n = \{g \in \Gamma \mid d_c(g, e) \leq n\}$  for  $n \in \mathbb{N}_0$ . The growth function of a Cayley graph  $\Gamma$  is defined as

$$\gamma(n) = |\Gamma_n|.$$

The asymptotic growth rate is defined as the limit

$$\rho = \lim_{n \rightarrow \infty} \gamma(n)^{1/n}.$$

**Proposition 2.5.2** *The limit  $\rho$  exists and  $\rho \geq 1$ . If the group is finite then  $\rho = 1$ . The following dichotomy holds. If two Cayley graphs of the same group are given, then the corresponding two limits are both equal to 1 or both strictly greater than 1.*

*Proof ([127])*

(a) We first show that the limit exists. Let  $\Gamma$  be a Cayley graph for some group  $G = \langle \tau_1, \dots, \tau_m \rangle$ , and  $\Gamma_l = \{g : d(g, e) \leq l\}$ . Since

$$\gamma(k+l) = |\{g \in \Gamma : d_c(g, e) \leq k+l\}| = \left| \bigcup_{g \in \Gamma_k} \Gamma_l \right| \leq \sum_{g \in \Gamma_k} \gamma(l) \leq \gamma(k)\gamma(l),$$

we find

$$\gamma(k) \leq \gamma(1)^k,$$

and thus

$$\limsup_{n \rightarrow \infty} \gamma(n)^{1/n} \leq \gamma(1).$$

We now fix  $l \in \mathbb{N}$ , and define  $k = \lfloor n/l \rfloor + 1$ . Then we have  $kl \geq n$ . Hence

$$\gamma(n) \leq \gamma(kl) \leq \gamma(l)^k = \gamma(l)^{\lfloor n/l \rfloor + 1},$$

and hence for any  $l \in \mathbb{N}$

$$\gamma(n)^{1/n} \leq \gamma(l)^{(\lfloor n/l \rfloor + 1)/n} \leq \gamma(l)^{(n/l + 1)/n} \leq \gamma(l)^{1/l + 1/n}.$$

Since this is true for any  $l \in \mathbb{N}$ , we obtain

$$\limsup_{n \rightarrow \infty} \gamma(n)^{1/n} \leq \gamma(l)^{1/l}$$

and hence

$$\limsup_{n \rightarrow \infty} \gamma(n)^{1/n} \leq \liminf_{l \rightarrow \infty} \gamma(l)^{1/l}.$$

Hence the limit exists and is not less than  $\gamma(1)$ .

- (b) Now we investigate the dependence of  $\rho$  on the Cayley graph for a given group. If the group is infinite, then  $\gamma(n+1) > \gamma(n)$ . Hence  $\gamma(n) \geq n$  and

$$\rho \geq \lim_{n \rightarrow \infty} n^{1/n} = 1.$$

Let  $G$  be generated by two sets of generators,  $G = \langle \tau_1, \dots, \tau_m \rangle = \langle \zeta_1, \dots, \zeta_l \rangle$  and let  $\Gamma^\tau, \Gamma^\zeta$  be the corresponding Cayley graphs. Let  $\gamma_\tau, \gamma_\zeta$  be the growth functions of  $\Gamma^\tau, \Gamma^\zeta$ , respectively.

Let  $\Gamma_n^\tau$  be the ball of radius  $n$  with center  $e$  with respect to the Cayley graph  $B^\tau$ , and  $\rho_\tau$  the corresponding limit, similarly define  $\gamma_\zeta, \Gamma_n^\zeta$ , and  $\rho_\zeta$  for  $\Gamma^\zeta$ .

Then there is  $k_0 \in \mathbb{N}$ , such that  $\zeta_i \in \Gamma_{k_0}^\tau, i = 1, \dots, l$ , i.e., each of the generators  $\zeta_i$  can be expressed as a word of generators  $\tau_i$  of length  $k_0$  or less. Thus, a word of length less or equal  $n$  (with respect to  $\Gamma^\zeta$ ) can be expressed as a word of length  $k_0 n$  or less in terms of  $\tau_i$ . Hence  $\gamma_\zeta(n) \leq \gamma_\tau(k_0 n)$  and

$$\rho_\zeta \leq \rho_\tau^{k_0}.$$

Similarly, there is a  $k_1$  such that  $\rho_\tau \leq \rho_\zeta^{k_1}$ .

□

Notice that the proof indicates that the same group, if it is exponentially growing, may have—in dependence on the choice of the generators—different asymptotic growth rates. All these asymptotic growth rates are larger than one. This proposition gives rise to the following definition.

**Definition 2.5.3** Let  $G$  be an infinite, finitely generated group for which  $\rho > 1$  holds for one Cayley graph. Then  $G$  is called a **group of exponential growth**. If there is a **polynomial  $p(n)$  such that  $\gamma(n) \leq p(n)$** , then it is called a **group of polynomial growth**. If the group is not of exponential nor of polynomial growth, then it is called of **intermediate growth**.

*Remark 2.5.4* There **are groups of all three growth types**. The existence of groups of intermediate growth has been proved as late as 1984 [78].

*Example 2.5.5* Let  $\gamma^{(n)}(l)$  be the growth function for  $F_n$ , the free group over  $n$  symbols,  $n > 1$ . First of all,  $\gamma^{(n)}(l)$  for  $l = 0, 1$  is given by

$$\gamma^{(n)}(0) = 1, \quad \gamma^{(n)}(1) = 2n + 1.$$

$\Gamma_{l+1}$  is the disjoint union of  $\Gamma_l$  and  $\Gamma_{l+1} \setminus \Gamma_l$ . Each vertex in  $\Gamma_l \setminus \Gamma_{l-1}$  has exactly  $2n-1$  neighbors in  $\Gamma_{l+1} \setminus \Gamma_l$ . As two different vertices in  $\Gamma_l \setminus \Gamma_{l-1}$  only have common neighbors in  $\Gamma_{l-1}$  we obtain  $|\Gamma_{l+1} \setminus \Gamma_l| = (2n-1) |\Gamma_l \setminus \Gamma_{l-1}|$ . Therefrom we find the recursion

$$\begin{aligned} \gamma^{(n)}(l+1) &= |\Gamma_l| + |\Gamma_{l+1} \setminus \Gamma_l| = \gamma^{(n)}(l) + (2n-1)|\Gamma_l \setminus \Gamma_{l-1}| \\ &= \gamma^{(n)}(l) + (2n-1)(\gamma^{(n)}(l) - \gamma^{(n)}(l-1)) \\ &= 2n \gamma^{(n)}(l) - (2n-1) \gamma^{(n)}(l-1). \end{aligned}$$

Hence,

$$\begin{pmatrix} \gamma^{(n)}(l+1) \\ \gamma^{(n)}(l) \end{pmatrix} = \begin{pmatrix} 2n & -(2n-1) \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \gamma^{(n)}(l) \\ \gamma^{(n)}(l-1) \end{pmatrix}.$$

The eigenvalues of this matrix are  $\lambda_- = 1$  resp.  $\lambda_+ = 2n-1$ , and the corresponding eigenvectors read

$$x_- = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad x_+ = \begin{pmatrix} 2n-1 \\ 1 \end{pmatrix}.$$

As

$$\begin{pmatrix} \gamma^{(n)}(1) \\ \gamma^{(n)}(0) \end{pmatrix} = \begin{pmatrix} 2n+1 \\ 1 \end{pmatrix} = \frac{n}{n-1}x_+ - \frac{1}{n-1}x_-$$

we find

$$A^l \begin{pmatrix} \gamma^{(n)}(1) \\ \gamma^{(n)}(0) \end{pmatrix} = \frac{n}{n-1}(2n-1)^l x_+ - \frac{1}{n-1}x_-.$$

This equation implies for  $l \in \mathbb{N}_0$

$$\gamma^{(n)}(l) = [n(2n-1)^l - 1]/(n-1).$$

The growth function  $\gamma(l)$  behaves asymptotically like  $\exp(\ln(2n-1)l)$ , i.e., the free groups over more than one symbol are groups of exponential growth with asymptotic growth rate  $2n-1$ .